25. Elimination Theory and Ideal Theory.

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In what follows I would like to show how elimination theory can be developed in exact parallel with the theory of zeros of polynomials in one indeterminate. This construction rests on a connection between Hentzelt's elimination theory, the methods of field theory as developed by Steinitz, and those of ideal theory as I myself developed it.

I first want to sketch the question in the case of polynomials in one indeterminate. As coefficient domain let a fixed field P be taken once for all, to which the concepts prime function, etc., refer. Here P may be assumed to be abstractly defined in the sense of Steinitz, but in the special case it may naturally coincide with the field of rational numbers or the field of all complex numbers – according to the assumption formerly usual in elimination theory. The theory of zeros for polynomials in one indeterminate now splits into four steps:

1. *The decomposition theorem*, which permits the reduction of the problem to one concerning prime functions – polynomials irreducible in P. For from

$$a(x) = p_1(x)^{\varrho_1} \cdots p_r(x)^{\varrho_r} = q_1(x) \cdots q_r(x)$$

– where the $p(x)$ denote distinct prime functions and the $q(x)$ therefore denote "greatest primary functions" – it follows that $a(x)$ vanishes at all and only the zeros of the prime functions $p(x)$.

2. *The construction of the field of zeros for a prime function* $p(x)$. Such a field of zeros $\mathfrak{R} = P(\alpha)$ – where α denotes a zero of $p(x)$ – can always, by Steinitz, be constructed as an extension field of P isomorphic to the residue-class field $(\mathfrak{R})$ of $p(x)$. Conversely, every field of zeros lying in an extension field of P is isomorphic to the residue-class field $(\mathfrak{R})$; field of zeros and residue-class field are thus abstractly identical.

3. *The decomposition of* $p(x)$ *into linear factors in the Galois field determined by* $p(x)$. By repeating the construction given under 2. one arrives at the essentially uniquely determined Galois field $P(\alpha_1, \ldots, \alpha_m)$, which is precisely sufficient for $p(x)$ to split into linear factors.

4. *The meaning of multiplicity for the original polynomial* $a(x)$. By 1., 2., 3., the question of the zeros, and the corresponding decomposition into linear factors, is settled for an arbitrary polynomial $a(x)$. The multiplicity may be regarded as the degree of the individual primary function $q(x)$ – that is, the number of residue classes modulo $q(x)$ linearly independent over P; for in the case of the algebraically closed coefficient domain P, where the $p(x)$ become linear, this number is precisely the multiplicity of the zeros.

I now pass to general elimination theory, and therefore take as underlying domain $\mathfrak{S}$, the domain of all polynomials in $x_1, \ldots, x_n$ with coefficients from P; elimination theory can then be defined as the *theory of the zeros of an ideal* in this polynomial domain. By an ideal $\mathfrak{a}$ I understand, as usual, a system of polynomials such that, together with f and g, it also contains $f - g$, and, together with f, it also contains Af, where A is an arbitrary polynomial in $\mathfrak{S}$. By a zero of $\mathfrak{a}$ I understand any tuple $(\alpha_1, \ldots, \alpha_n)$ in an extension field of P such that $f(\alpha) = 0$ for every polynomial f belonging to the ideal. If $f_1, \ldots, f_t$ is an ideal basis of $\mathfrak{a}$ (such a finite basis always exists), so that $f = A_1 f_1 + \cdots + A_t f_t$ holds for every f in $\mathfrak{a}$, then the zeros of $\mathfrak{a}$ are plainly identical with the common zeros of $f_1, \ldots, f_t$. Conversely, every finite system of polynomials can be regarded as a basis of the ideal generated by it. Thus the definition of zeros just given for elimination theory is identical with the customary one, but avoids the difficulty inherent in arbitrarily singling out a basis. For polynomials in one indeterminate, where only principal ideals are involved, this difficulty does not arise, since the basis polynomial is uniquely determined up to multiplication by a unit – that is, by a nonzero element of P – but even there the ideal-theoretic conception will give a more exact insight.

Elimination theory now splits, correspondingly to the case of one indeterminate, into four steps:

I. *The decomposition theorem* – the representation of an ideal by greatest primary components – which permits the reduction of the problem to one concerning prime ideals. Here an ideal is called prime if, whenever it divides a product, it divides at least one factor; it is called primary if, whenever it divides a product, it divides at least one factor or else a power of every factor. To every primary ideal $\mathfrak{q}$ there belongs one and only one associated prime ideal $\mathfrak{p}$, which is a divisor of $\mathfrak{q}$ and a power of which is divisible by $\mathfrak{q}$. One has the

[1)] A detailed presentation and references to the literature are in Math. Ann. 90 and in a forthcoming paper.

representation as least common multiple:

$$\mathfrak{a} = [\mathfrak{q}_1, \ldots, \mathfrak{q}_r]; \qquad \mathfrak{p}_1^{\sigma_1} \cdots \mathfrak{p}_r^{\sigma_r} \equiv 0\,(\mathfrak{a}).$$

Here the $\mathfrak{q}$ are greatest primary components; that is, each $\mathfrak{q}$ is primary, but the least common multiple of any two $\mathfrak{q}$'s is not primary. Moreover, without loss of generality it may be assumed that no $\mathfrak{q}$ divides the least common multiple of the remaining ones, so that no $\mathfrak{q}$ can simply be omitted; the $\mathfrak{p}$ are the associated prime ideals of the $\mathfrak{q}$. Now it is not the $\mathfrak{q}$, but – and herein lies the analogue to the case of one indeterminate – their associated prime ideals $\mathfrak{p}$ that are uniquely determined by $\mathfrak{a}$; and since each $\mathfrak{p}$ is a divisor of $\mathfrak{a}$, and conversely a product of powers of the $\mathfrak{p}$ is divisible by $\mathfrak{a}$, the zero set of the ideal is therefore the union of the zero sets of its associated prime ideals.

II. *The construction of the field of zeros for a prime ideal* $\mathfrak{p}$. By definition the residue classes modulo a prime ideal form a ring without zero divisors, containing at least one element distinct from the zero element as soon as $\mathfrak{p}$ is different from the unit ideal; by forming quotients this ring can therefore be extended to the residue-class field $(\mathfrak{R})$ of $\mathfrak{p}$. One can always construct an extension field $\mathfrak{R}$ of P isomorphic to $(\mathfrak{R})$ which becomes the field of zeros of $\mathfrak{p}$; that is, $\mathfrak{R} = P(\alpha_1, \ldots, \alpha_n)$, where $(\alpha_1, \ldots, \alpha_n)$ is a zero of $\mathfrak{p}$. Here $\mathfrak{R}$ has transcendence degree k $(0 \leq k < n)$ over P, hence contains k elements algebraically independent over P, whereas any $(k+1)$ elements become algebraically dependent over P. Conversely, every field of zeros of transcendence degree k lying in an extension field of P is isomorphic to the residue-class field $(\mathfrak{R})$; field of zeros of transcendence degree k and residue-class field are therefore abstractly identical.

III. *Decomposition of the norm of* $\mathfrak{p}$ *into linear factors in the Galois field determined by* $\mathfrak{p}$. If the indeterminates $x_1, \ldots, x_n$ are regarded as generated from original indeterminates $y_1, \ldots, y_n$ by a linear transformation with indeterminates as coefficients, then the field of zeros $\mathfrak{R}$ of transcendence degree k can be regarded as a finite algebraic extension field of $P(x_{i+1}, \ldots, x_n)$ $(k = n - i)$. One can therefore pass to the Galois field $\overline{\mathfrak{R}}$ over $P(x_{i+1}, \ldots, x_n)$, in which the prime function $P^{(i)}(x_i, x_{i+1}, \ldots, x_n)$ with the zero α_i splits into linear factors. This prime function $P^{(i)}$, when one returns to the original indeterminates y, plays the role of the fundamental equation; for α_i becomes equal to the fundamental form $u_1\beta_1 + \cdots + u_n\beta_n$, where $\beta = (\beta_1, \ldots, \beta_n)$ is a zero tuple in the y-variables that depends algebraically on the parameters $x_{i+1}, \ldots, x_n$. The decomposition into linear factors thus gives the product over all zeros. But the prime function $P^{(i)}$, or a power of it, may also be regarded as the norm $N(\mathfrak{p})$ of $\mathfrak{p}$, by considering $\mathfrak{p}$ as a module of linear forms in the power products of $x_1, \ldots, x_{i-1}$, taking $P(x_{i+1}, \ldots, x_n)$ instead of P as coefficient domain, and defining the norm as the product over all elementary divisors – of which only finitely many are distinct from the unit. For a perfect field P as coefficient domain one always obtains $N(\mathfrak{p}) = P^{(i)}$; for an imperfect field a power of $P^{(i)}$ may occur as the norm.

IV. *The norm of the original ideal and the meaning of multiplicity.* If correspondingly one regards the ideal $\mathfrak{a}$ successively as a module of linear forms in the power products of $x_1, \ldots, x_{i-1}$ $(i = 1, 2, \ldots, n)$, then it can be shown that, taking $P(x_{i+1}, \ldots, x_n)$ as coefficient field, this module has in each case only finitely many elementary divisors distinct from the unit; this is a finiteness condition following from the existence of an ideal basis. The product over these finitely many elementary divisors gives the individual norms; the norm $N(\mathfrak{a})$ is defined as the product of all individual norms. The greatest primary factors $Q^{(i)}(x_i, x_{i+1}, \ldots, x_n)$ of $N(\mathfrak{a})$ correspond one-to-one to the associated prime ideals $\mathfrak{p}$ of $\mathfrak{a}$, in such a way that $Q^{(i)}$ becomes a power of $P^{(i)}$, where $P^{(i)}$ (or, if necessary, a power of $P^{(i)}$) denotes the norm $N(\mathfrak{p})$ of $\mathfrak{p}$. Thus there arises a decomposition

$$N(\mathfrak{a}) = N(\mathfrak{p}_1)^{\varrho_1} \cdots N(\mathfrak{p}_r)^{\varrho_r} = Q_1 \cdots Q_r,$$

– where if necessary $N(\mathfrak{p})$ is to be replaced by the highest elementary divisor $E(\mathfrak{p}) = P^{(i)}$ – and thus, by I, II, III, the decomposition of the norm into linear factors has been effected for all zeros of $\mathfrak{a}$. The multiplicity of the zeros corresponding to the individual prime ideal $\mathfrak{p}$ of $\mathfrak{a}$ can be regarded as the degree of $Q^{(i)}$, which can be interpreted as the number of residue classes linearly independent over $P(x_{i+1}, \ldots, x_n)$. More precisely, these are the residue classes modulo certain isolated components of the decomposition uniquely determined by $\mathfrak{p}$ and by the prime ideals of higher dimension: namely, the residue classes of the fundamental ideal of the $(i-1)$-st stage modulo the component, determined by $\mathfrak{p}$, of the fundamental ideal of the i-th stage. The latter component has $\mathfrak{p}$ and all prime ideals of higher dimension as associated prime ideals, while the former has only the prime ideals of higher dimension.

A complete *integration of the case of one indeterminate* is obtained when one passes here too to the principal ideal. The decomposition given under 1. then splits – according to whether $a(x)$ is regarded as basis polynomial or norm – into two separate decompositions: into the representation of the principal ideal as the least common multiple of greatest primary components according to I, which here amounts to a representation as a product of powers of prime ideals; or into the decomposition of the norm as a product of powers of the norms of the individual prime ideals according to IV. In the construction of the field of zeros according to 2., what is in question in reality is the field of zeros of the prime ideal defined by $p(x)$, while in 3. the norm is decomposed into linear factors. The definition of multiplicity according to 4. is subsumed under that given in IV, since the fundamental ideal of the zeroth stage is equal to the unit ideal, while for one indeterminate the fundamental ideal of the first stage already equals the ideal itself.

The *integration into the usual elimination theory* is given by the fact that the norm of the ideal has a resultant interpretation. In this interpretation, the old problem of elimination theory thus becomes a part of the ideal theory of the polynomial domain, namely that part which, beside the decomposition theorems for ideals, also treats the decomposition of their norms and the connection between the two decompositions.

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